Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: MEHRSHINE's 50 KW at Nuzvid by Vaseem

Variant: MEHRSHINE LLP

Tables on a building

System power: 23.94 kWp

Nuzvid - India

Author

PVsyst V7.4.7
VC0, Simulation date:
25/02/26 20:00
with V7.4.7

Project summary

Geographical Site
Nuzvid
India

Situation

Latitude	16.79 °N
Longitude	80.85 °E
Altitude	106 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Nuzvid
Meteonorm 8.1 (1996-2015), Sat=100% - Synthetic

System summary

Grid-Connected System

Tables on a building

PV Field Orientation
Fixed plane
Tilt/Azimuth 16 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	126 units
Pnom total	23.94 kWp

Inverters

Nb. of units	1.5 units
Pnom total	22.50 kWac
Pnom ratio	1.064

Results summary

Produced Energy	31268 kWh/year	Specific production	1306 kWh/kWp/year	Perf. Ratio PR	72.79 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Near shading definition - Iso-shadings diagram 4
Main results 5
Loss diagram 6
Predef. graphs 7
P50 - P90 evaluation 8

General parameters

Grid-Connected System

Tables on a building

PV Field Orientation

Orientation		**Sheds configuration**		**Models used**	
Fixed plane		Nb. of sheds	24 units	Transposition	Perez
Tilt/Azimuth	16 / 0 °	**Sizes**		Diffuse	Perez, Meteonorm
		Sheds spacing	2.20 m	Circumsolar	separate
		Collector width	1.63 m		
		Ground Cov. Ratio (GCR)	74.0 %		
		Shading limit angle			
		Limit profile angle	35.3 °		

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	ZNshine Solar
Model	ZXM5-72-190/MS
(Original PVsyst database)	
Unit Nom. Power	190 Wp
Number of PV modules	126 units
Nominal (STC)	23.94 kWp
Modules	9 string x 14 In series
At operating cond. (50°C)	
Pmpp	21.41 kWp
U mpp	456 V
I mpp	47 A

Inverter

Manufacturer	GESolar
Model	GES-15KTL
(Original PVsyst database)	
Unit Nom. Power	15.0 kWac
Number of inverters	3 * MPPT 50% 1.5 units
Total power	22.5 kWac
Operating voltage	400-850 V
Pnom ratio (DC:AC)	1.06
No power sharing between MPPTs	

Total PV power

Nominal (STC)	24 kWp
Total	126 modules
Module area	161 m²
Cell area	138 m²

Total inverter power

Total power	22.5 kWac
Nb. of inverters	2 units
	0.5 unused
Pnom ratio	1.06

Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 163 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
ASHRAE Param.: IAM = 1 - bo (1/cosi -1)
bo Param. 0.05

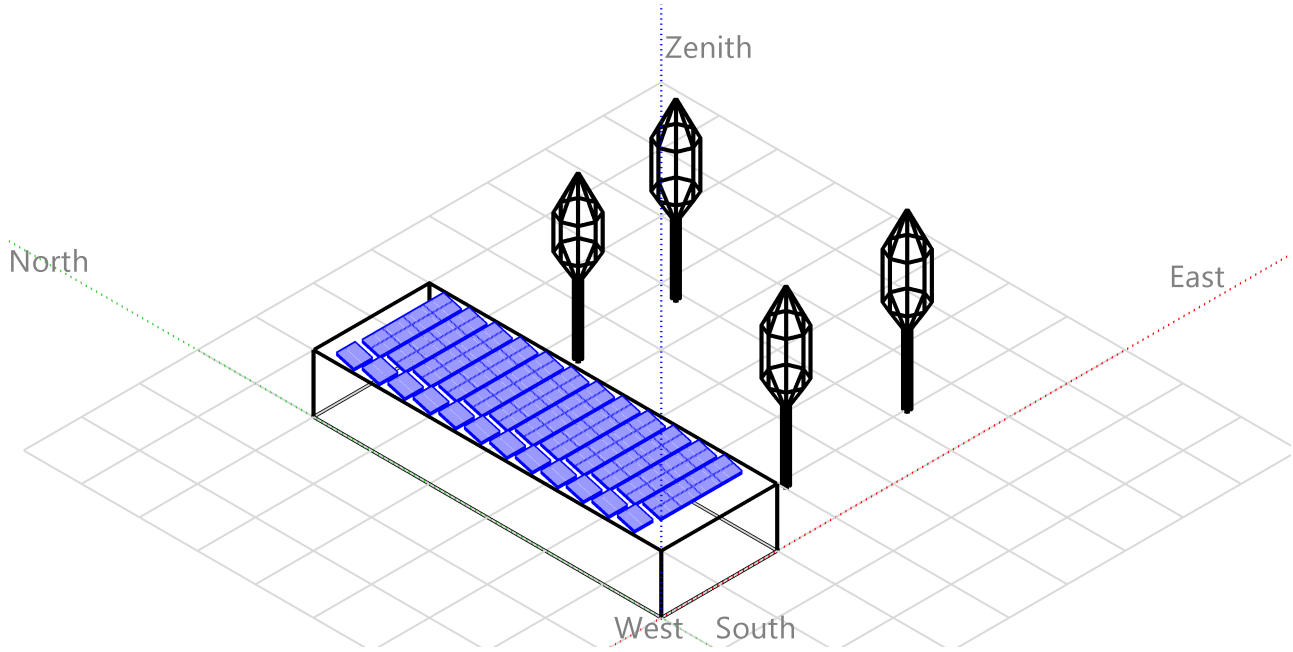
System losses

Unavailability of the system
Time fraction 2.0 %
7.3 days,
3 periods

Auxiliaries loss

Near shadings parameter

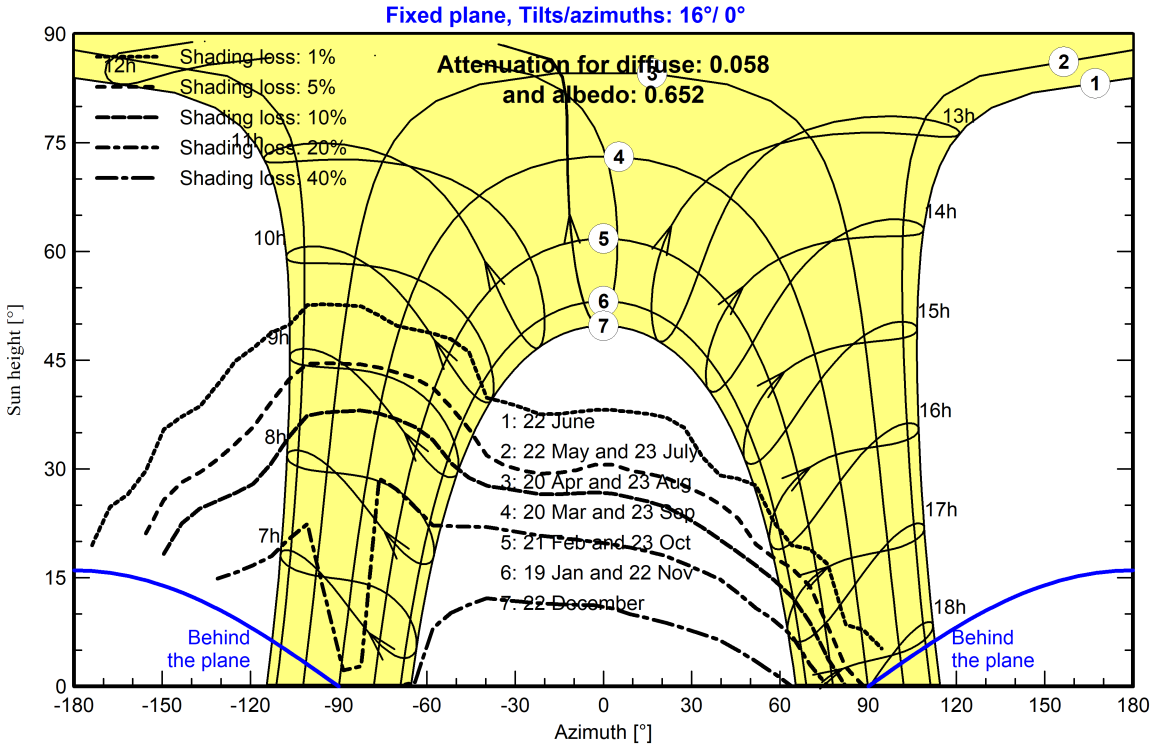
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 16°/ 0°

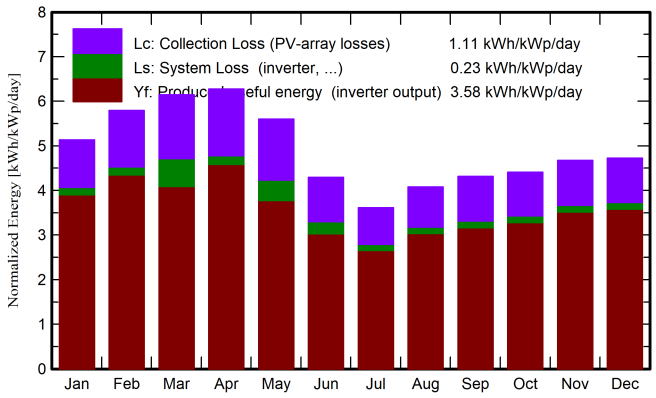


Main results

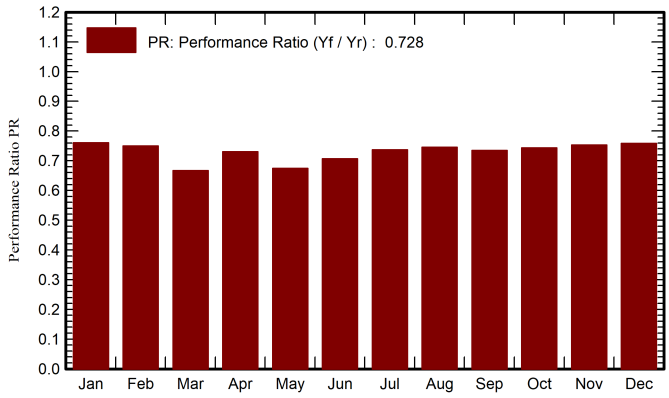
System Production

Produced Energy	31268 kWh/year	Specific production	1306 kWh/kWp/year
		Perf. Ratio PR	72.79 %

Normalized productions (per installed kWp)



Performance Ratio PR



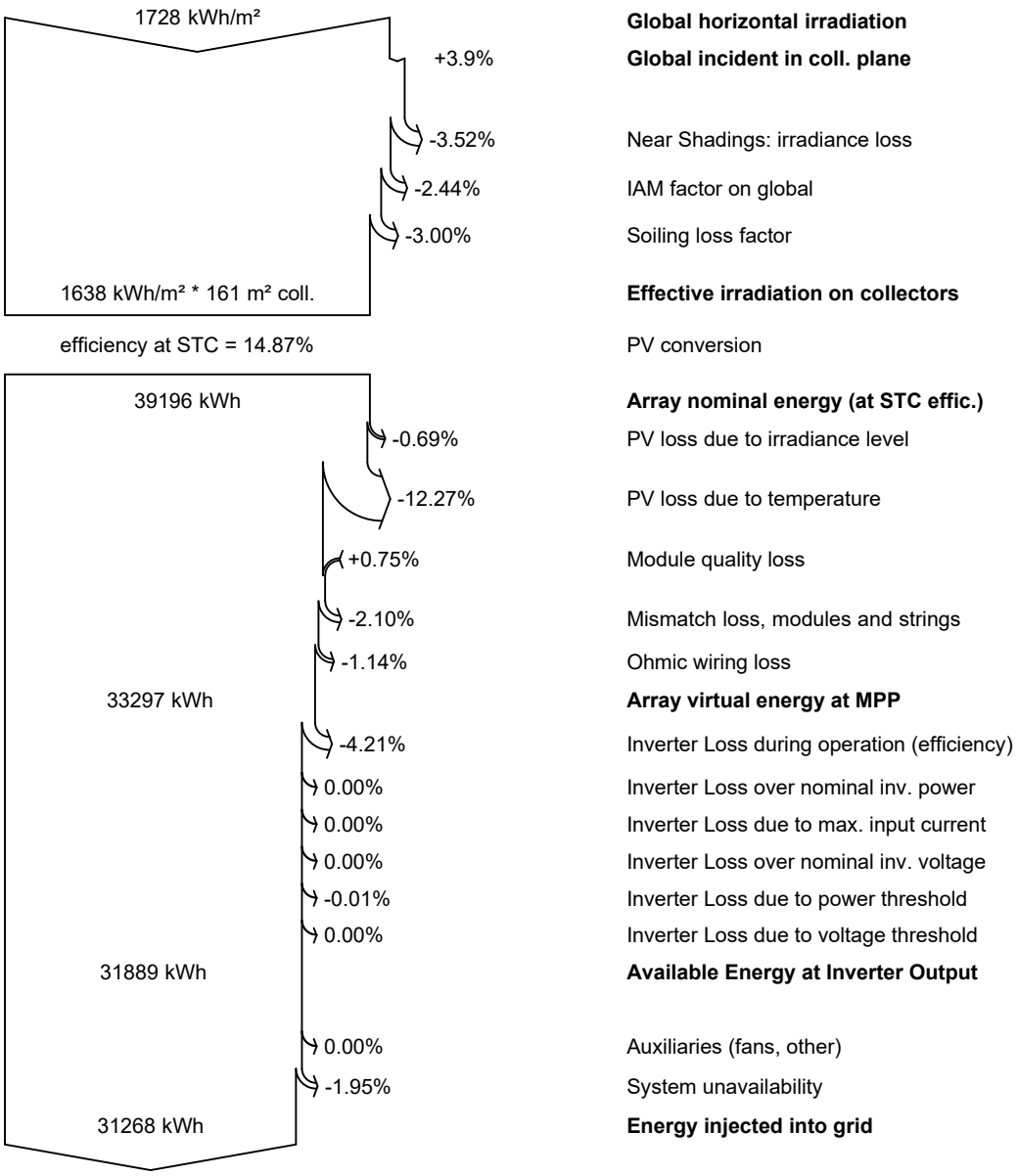
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	137.3	64.75	23.20	159.2	145.2	3022	2900	0.761
February	145.9	67.18	25.21	162.4	148.7	3037	2916	0.750
March	181.3	79.83	28.36	190.6	175.3	3501	3041	0.667
April	189.3	83.23	30.67	188.3	173.7	3435	3295	0.731
May	182.9	95.14	33.17	173.6	159.3	3147	2804	0.674
June	137.1	88.99	31.56	128.8	117.0	2374	2180	0.707
July	118.3	84.94	30.24	111.9	100.9	2075	1975	0.737
August	130.0	92.45	29.26	126.6	114.7	2365	2258	0.745
September	127.6	69.77	28.43	129.5	117.9	2386	2277	0.734
October	128.7	76.10	27.74	136.7	124.6	2546	2435	0.744
November	123.8	60.67	25.38	140.3	128.0	2637	2527	0.753
December	125.6	60.91	23.66	146.6	133.1	2772	2660	0.758
Year	1727.9	923.95	28.09	1794.4	1638.3	33296	31268	0.728

Legends

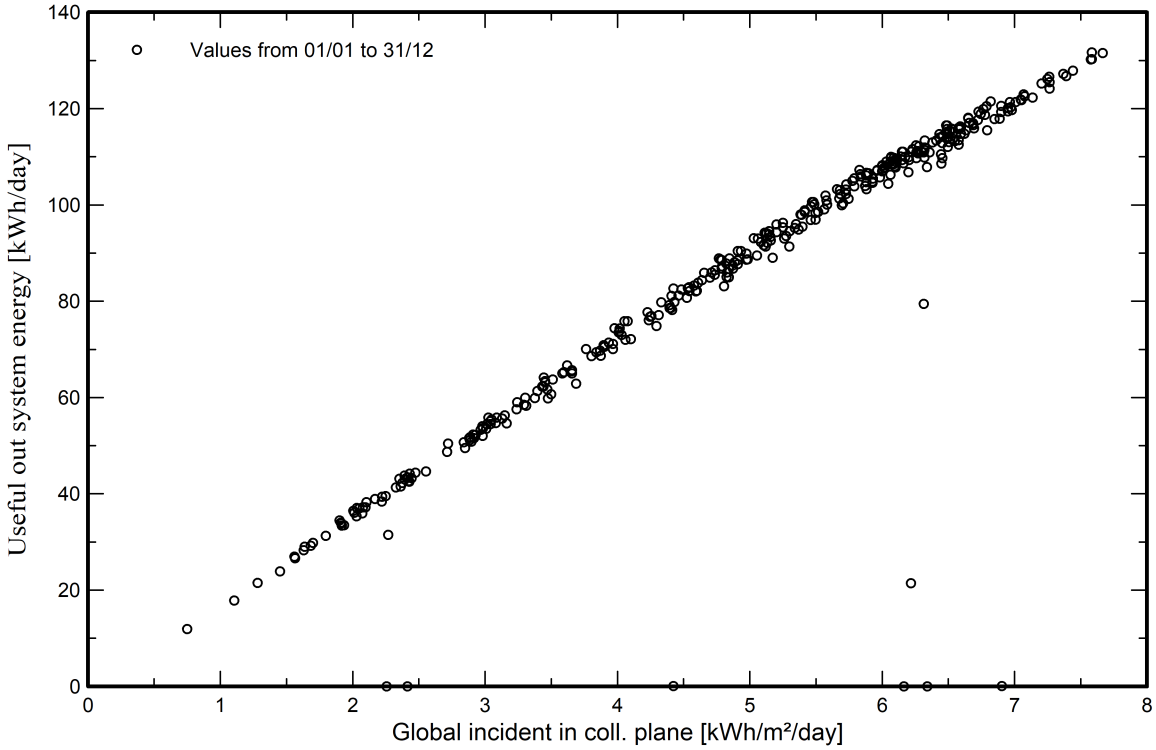
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

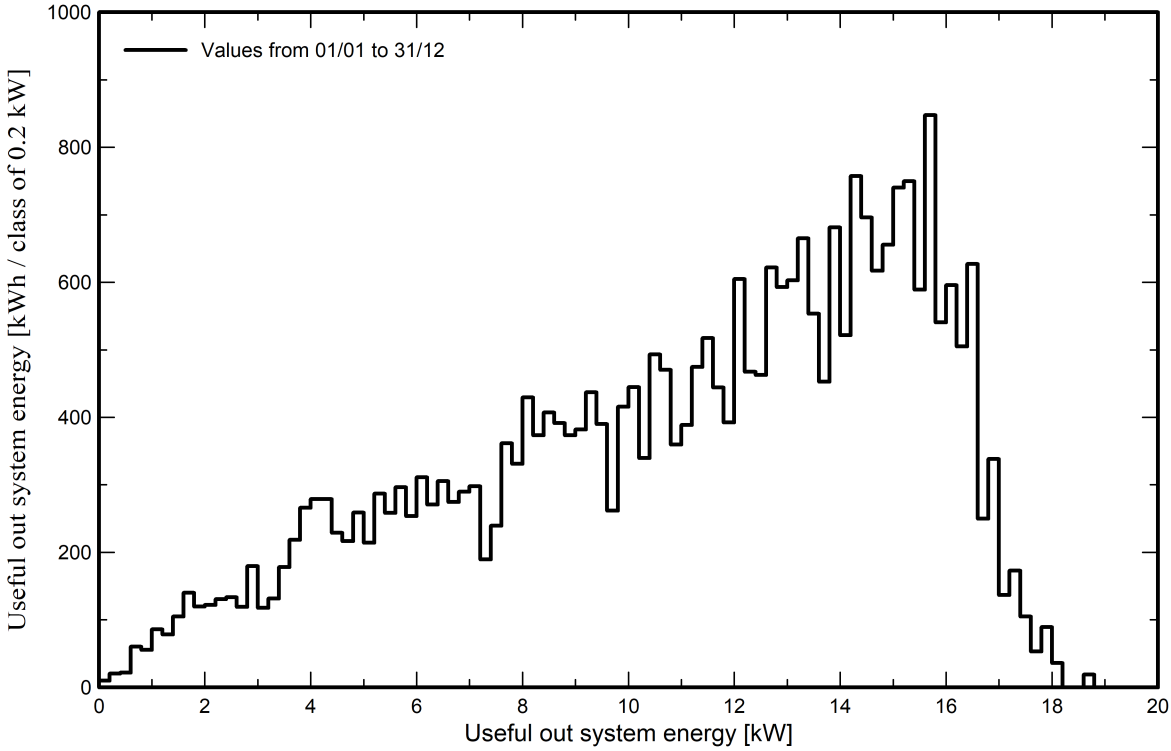


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 8.1 (1996-2015), Sat=100%
Kind	Not defined
Year-to-year variability(Variance)	0.0 %
Specified Deviation	

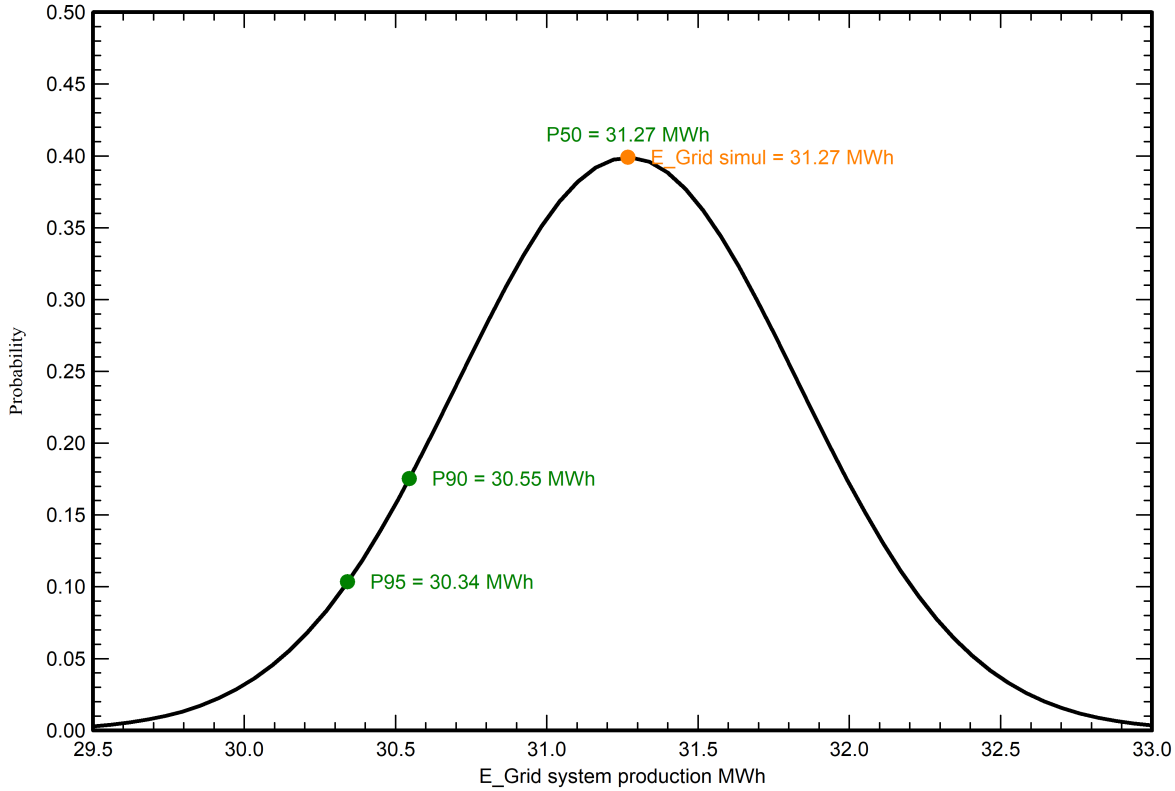
Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	1.8 %

Annual production probability

Variability	0.56 MWh
P50	31.27 MWh
P90	30.55 MWh
P95	30.34 MWh

Probability distribution